Reflection-Guided Dynamics Rectification for Text-to-Image Generation

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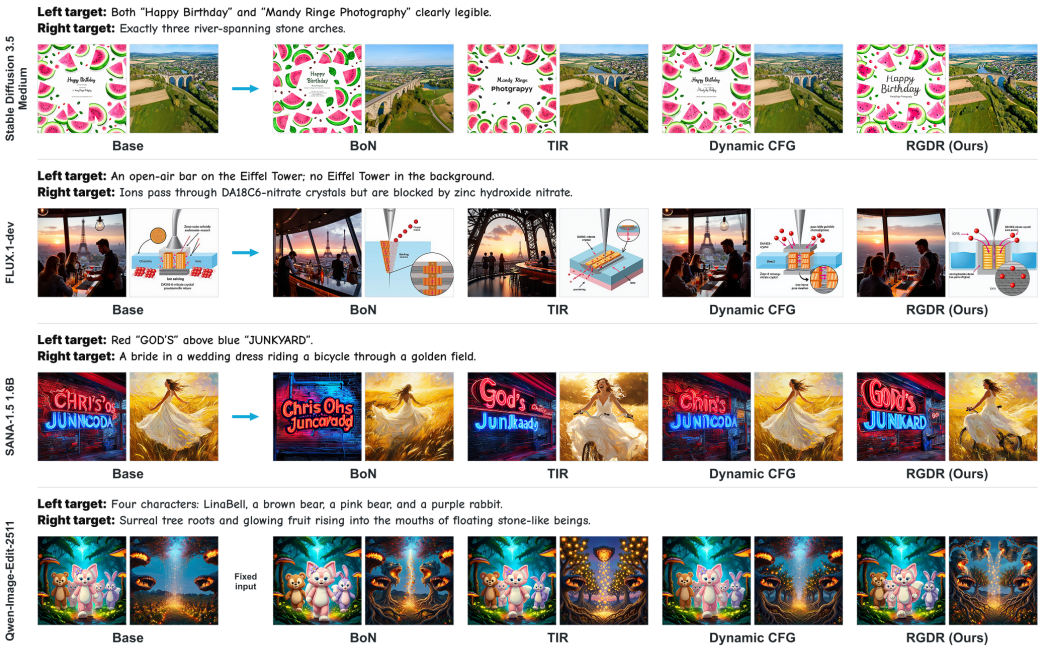

Figure 1: **Reflection-guided Dynamics Rectification (RGDR).** RGDR turns reflections on existing outputs into stepwise corrections along sampling trajectories. Qualitative comparisons with BoN, TIR, and Dynamic CFG across one image-editing model and three text-to-image models illustrate how different methods address residual errors and affect existing content during rectification.

ABSTRACT

Fine-grained rectification of existing generations under a limited additional budget remains a key problem in test-time optimization (TTO) for image generation. As for prior approaches, they could identify residual errors, but could not directly ensure that the model acts on the feedback during reasoning. Such errors stem from multiple sources, such as the initial noise or text prompt, and ultimately induce an execution gap of failing to handle potential deviations along the trajectory generating. However, classic models generally cannot remedy this gap. To narrow it, we propose Reflection-Guided Dynamics Rectification (RGDR), which turns reflections on completed outputs into corrective interventions on the process

Interactive demos, dataset, and code for this paper are available at: https://mair-lab-hust.github.io/RGDR/.

of sampling dynamics at the trajectory level. Specifically, RGDR could impact every sampling step by adapting to instantaneous intervention. Moreover, RGDR generates flexible candidates while adjudicating between them and the reference. As experiments show, rectification from RGDR starts from the existing base trajectory and reflection, and RGDR is substantially more effective than former trajectory exploration. Under matched budgets for additional image generation, RGDR's gain in the cross-model average final checklist score over the reference is $1.51\times$ that of TIR prompt refinement and $1.81\times$ that of Dynamic CFG.

1 INTRODUCTION

Research on text-to-image generation is increasingly directed toward achieving superior quality and better alignment with prompts. Specifically, as inference costs accumulate, test-time optimization (TTO) demonstrates its superiority without updating the parameters of a pretrained model. As for prior methods, they have made progress through sampling, search, and feedback-driven refinement, using generative models to produce additional candidates or image-editing models to revise the output (Wan et al., 2026; Jiang et al., 2026). However, a challenge has arisen in optimizing outputs under a limited additional budget, correcting residual errors while preserving prior progress.

We classify the methods commonly into two types. One is the noise-level scaling, and the other is the prompt-level scaling. These are two common approaches that improve the quality of image generation by searching for noise initialization and using the text prompt respectively. (Zhuo et al., 2025). Although the previous approaches are still effective, it is no longer possible to continuously achieve higher gains by further adjusting the noise or prompts when the budget is limited. Figure 2 contrasts the behavior of Noise Sampling and Prompt Refinement. The former produces substantial visual variation, whereas the latter repeatedly produces similar outputs, yet key requirements remain unmet in the illustrated examples. Across the evaluated settings, the mean score gain per additional image also generally decreases as the budget grows. Therefore, we believe that the quality of noise initialization and text guidance do indeed affect the correction effect, but the real bottleneck lies in the fact that the way the model transforms the guidance into images through continuous state updates along the generation trajectory is rather fragile. (Ho et al., 2020). If the intended correction does not succeed through updates, residual errors can persist, leaving an execution gap between the intention of the correction and the generative response. This motivates trajectory-level methods, which intervene in the sampling process rather than only searching over its initial conditions.

In terms of prior work, they have explored TTO along generation trajectories, using evaluations or diagnoses of intermediate states to guide subsequent sampling (Ma et al., 2025; Papalampidi et al., 2025; Lv et al., 2025). However, these trajectory-level TTO methods do not fully exploit the information available in follow-up rectification, such as the completed reference revealing residual errors. Therefore, we consider converting the residual-error diagnosis of a completed output into guidance that steers a new trajectory while retaining the reference configuration. Related work provides a conceptual basis for such interventions, as SEGA (Brack et al., 2023) introduces additional semantic guidance, while CFG-Ctrl (Wang et al., 2026a) interprets velocity-field modulation from a control-theoretic perspective.

Given the above analysis, proposing a trajectory-level test-time scaling method for rectifying existing generations is crucial. Therefore, we introduce RGDR, which adds a reflection-guided correction field to the base velocity field at each sampling step and generates complete candidates by controlling the correction strength. The correction field is the difference between velocity predictions conditioned on the desired state (Target) and the current erroneous state (Reference). Under suitable regularity conditions, we establish continuity and bounded deviation of candidates and derive sufficient conditions for corrective progress to outweigh accompanying interference. Figure 3 illustrates how feedback correction acts throughout generation by comparing trajectories from the same initial noise.

To show the superiority in reasoning tasks, we evaluate RGDR on tasks drawn from PRISM-Bench (Fang et al., 2025) and SciIR-Bench (Ma et al., 2026) using three text-to-image models, Stable Diffusion 3.5 Medium, SANA-1.5 1.6B, and FLUX.1-dev, and the image-editing model Qwen-Image-Edit-2511. For each task and model, all methods continue from the same reference selected by preceding optimization. Under matched budgets for additional image generation, we

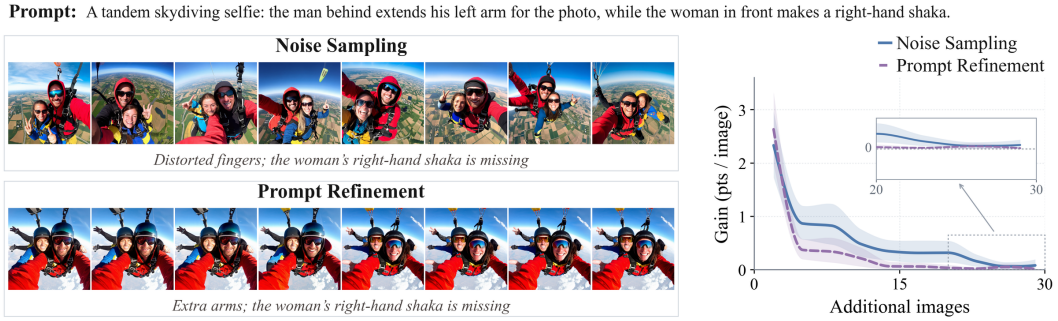

Figure 2: **Limitations of noise sampling and prompt refinement in follow-up rectification.** The left panel shows qualitative results from noise sampling (top) and prompt refinement (bottom). The right panel reports mean score gain per additional image across generation budgets.

compare RGDR with BoN, TIR, and Dynamic CFG, which respectively resample the noise, refine the prompt, and tune the guidance scale. RGDR achieves the highest average final checklist score and repair rate across all four models, providing greater rectification gains than further noise or prompt adjustments and than adjusting native guidance strength alone. The same advantage on previously edited outputs shows that RGDR can serve as an additional rectification mechanism within existing generation and editing workflows. Direction ablations show the advantage of the Target–Reference construction over the tested alternatives, while strength analysis shows that effective correction depends on intervention direction and magnitude, rather than larger perturbations alone.

2 Background and Related Work

2.1 Flow-Based Generation

We formulate trajectory rectification in flow models as an intervention on the sampling velocity field. Flow models describe continuous transformations between noise and data distributions through time-dependent velocity fields. Flow matching (Lipman et al., 2023) and rectified flow (Liu et al., 2023) learn these fields and generate samples using numerical ordinary differential equation (ODE) solvers. Under the time convention from the noise endpoint T to the data endpoint 0, a sampling trajectory conditioned on c satisfies

$$\frac{\mathrm{d}x_t}{\mathrm{d}t} = V(t, x_t; c), \qquad x_T \sim \mathcal{N}(0, I_d), \qquad t : T \to 0. \tag{1}$$

Here, $x_t \in \mathbb{R}^d$ is the generative state, $T > 0$ is the sampling horizon, I_d is the d-dimensional identity matrix, and $V(t, x; c) \in \mathbb{R}^d$ is the velocity field incorporating the chosen guidance configuration. The states x_T and x_0 denote the initial noise and generated endpoint, respectively. In latent-space models, a decoder maps x_0 to the final image (Rombach et al., 2022). This framework supports both text-to-image generation and image editing. For example, FLUX.1 Kontext (Black Forest Labs et al., 2025) unifies image generation and conditional editing within a flow-matching model.

2.2 Semantic Guidance and Additive Velocity Fields

Semantic guidance combines or modifies model predictions to change state updates during sampling, thereby steering generation trajectories. Classifier-free guidance (CFG) (Ho & Salimans, 2022) combines conditional and unconditional predictions and scales their difference. SEGA (Brack et al., 2023) introduces independently adjustable concept directions alongside the original prompt guidance, allowing additional semantics to influence sampling updates.

RGDR constructs reflection-guided trajectory corrections using the additive velocity-field formulation

$$V_\alpha(t, x) = V_{\text{base}}(t, x) + \alpha D(t, x), \tag{2}$$

where $V_{\text{base}}(t, x)$ is the base velocity field, $D(t, x) \in \mathbb{R}^d$ is an additional semantic direction field, and α is a scaling coefficient. Both fields are evaluated at the same time and current state.

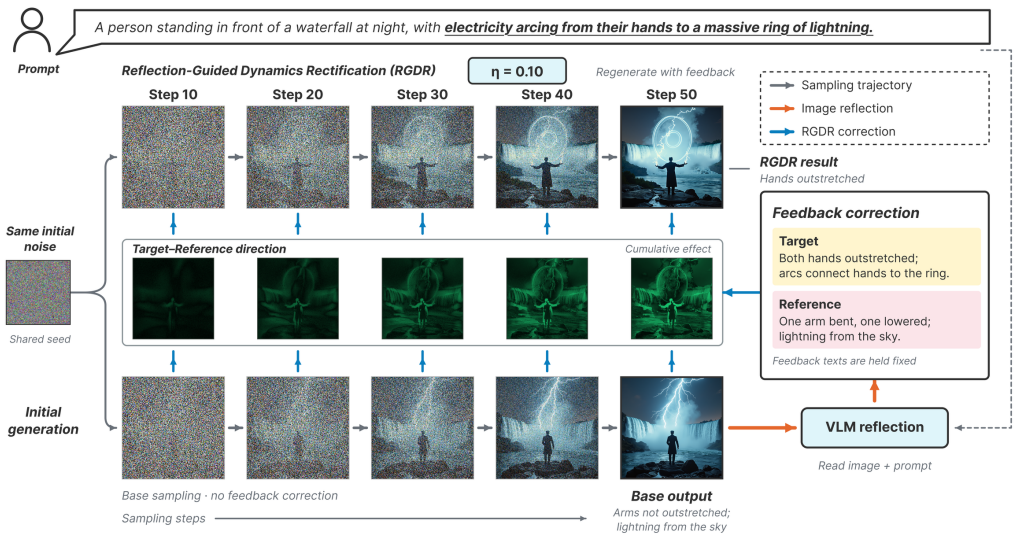

Figure 3: **Stepwise visualization with RGDR.** Reflections on the completed reference provide fixed Target–Reference conditions for constructing the correction field. Heatmaps show the spatial distribution of the correction signal along the generation trajectory.

2.3 Test-Time Optimization for Image Generation

Test-time optimization improves image generation through additional sampling, search, and feedback-driven refinement (Wan et al., 2026). BoN (Ma et al., 2025) selects candidates generated from different noise initializations, while TIR (Khan et al., 2025) refines prompts using the current image and accumulated feedback. ReflectionFlow (Zhuo et al., 2025) and Reflect-DiT (Li et al., 2025) train models to revise outputs using existing images and textual reflections, whereas RAISE (Jiang et al., 2026) coordinates prompt rewriting, noise resampling, and image editing without additional training. These methods use candidate selection or feedback-driven correction to organize subsequent generation and editing around existing outputs.

Trajectory-Level Scaling. Related work uses evaluations and diagnoses of intermediate states to adjust generation trajectories. Search over Paths (Ma et al., 2025) evaluates denoised predictions to select subsequent branches, while Dynamic CFG (Papalampidi et al., 2025) uses latent-space evaluators to select guidance scales at each step. PPAD (Lv et al., 2025) diagnoses intermediate previews and revises subsequent denoising through corrective conditions and rollback. T2S2 (Sakurai et al., 2026) detects prohibited concepts in intermediate clean-image estimates, optimizes a low-rank conditioning residual, and restarts sampling.

3 Method

3.1 Controlled Superposition of the Correction Field

Continuity and bounded deviation of the candidate family. Starting from Eq. (2) in Section 2.2, fix the base field, correction field, and initial noise, and consider

$$\mathrm{d}x_t = [V_{\text{base}}(t, x_t) + \alpha D(t, x_t)]\,\mathrm{d}t, \qquad t : T \to 0. \tag{3}$$

We omit the dependence of the trajectory state on α and denote the endpoint of a complete generation by $\gamma_D(\alpha)$. A strength range $\mathcal{A}$ defines the candidate family $\Gamma_D(\mathcal{A}) = \{\gamma_D(\alpha) : \alpha \in \mathcal{A}\}$, with reference endpoint $x^\star = \gamma_D(0)$.

Theorem 3.1 (Continuity and bounded deviation of the candidate family)**.** *Fix $r > 0$. Suppose that the dynamics admit a unique solution for every $\alpha \in [0, r]$. On a common region containing the relevant trajectories, assume that V_{base} and D are continuous in time and uniformly Lipschitz in the state, and that D is bounded. Then there exists a finite constant K_r, independent of α, α', such that,*

for all $\alpha, \alpha' \in [0, r]$,

$$\|\gamma_D(\alpha) - \gamma_D(\alpha')\| \le K_r|\alpha - \alpha'|, \qquad \|\gamma_D(\alpha) - x^\star\| \le K_r\alpha. \tag{4}$$

The two bounds describe continuous dependence of the endpoint on strength and its deviation from the reference, respectively. Candidates converge to the reference endpoint as $\alpha \to 0$. The proof is provided in Appendix B.1.

Relative reparameterization of the intervention strength. The theorem characterizes endpoint variation for a fixed direction field, but the relative effect of the raw coefficient α still depends on the norms of the correction and base fields. We therefore reparameterize the additional term relative to the base velocity norm,

$$\begin{aligned} \overline{D}(t,x) &= \frac{\|V_{\text{base}}(t,x)\|}{\|D(t,x)\| + \varepsilon} D(t,x), \qquad \varepsilon > 0, \\ V_{\eta,s}(t,x) &= V_{\text{base}}(t,x) + \eta s(t) \overline{D}(t,x). \end{aligned} \tag{5}$$

Here, $\eta \ge 0$ sets the overall relative strength and $s(t) \ge 0$ is a fixed, bounded temporal schedule. By construction,

$$\|V_{\eta,s}(t,x) - V_{\text{base}}(t,x)\| \le \eta s(t) \|V_{\text{base}}(t,x)\|. \tag{6}$$

The correction at each sampling step is thus scaled relative to the current base velocity, with $\eta s(t)$ bounding its relative norm. This gives η a dimensionless interpretation as a relative intervention parameter, rather than merely a multiplier of the raw direction.

Temporal scheduling. The schedule $s(t)$ distributes the correction strength along the generation trajectory. Under the convention $t : T \to 0$, $s(t) = 1$ gives constant intervention, $s(t) = t/T$ places greater strength early in sampling, and $s(t) = 1 - t/T$ places greater strength later. Other nonnegative bounded schedules can also be used. Model-specific schedules and scheduling experiments are provided in Appendices C.2, D.1, and D.2.

With the direction field and schedule fixed, let $\gamma(\eta)$ denote the endpoint of a complete generation under $V_{\eta,s}$. Applying Theorem 3.1 with direction field $s\overline{D}$ and strength η, under the corresponding regularity conditions, shows that candidate endpoints vary continuously with η, have bounded deviations from the reference, and converge to the reference endpoint as $\eta \to 0$. Figure 4 illustrates this candidate space continuously parameterized by relative strength.

3.2 Semantic Construction of the Correction Field

Corrective progress and accompanying interference. Fix the task q, feedback F, and the base field, initial noise, and schedule s from Section 3.1. We write D_F for the direction field D constructed from F, and retain $\gamma(\eta)$ for the endpoint under relative reparameterization. Let $\Phi_q : \mathbb{R}^d \to \mathbb{R}^{m_q}$ be a fixed task representation used for analysis, where m_q is its dimension. For latent-space generation, this mapping includes image decoding. Define

$$z(\eta) = \Phi_q(\gamma(\eta)), \qquad \Delta z(\eta) = z(\eta) - z(0). \tag{7}$$

In this representation space, let the unit vector $e_{\text{fix}} \in \mathbb{R}^{m_q}$ specify the desired correction direction, and let $P_{\text{keep}} \in \mathbb{R}^{m_q \times m_q}$ be the orthogonal projector onto the local subspace sensitive to properties already satisfied by the reference, with $P_{\text{keep}}^\top = P_{\text{keep}}^2 = P_{\text{keep}}$. Let $\beta > 0$ weight the interference term. These analytical objects remain fixed for the current correction and are assumed to capture the progress and interference of interest. Define the net corrective progress

$$\Delta_q(\eta) = \langle e_{\text{fix}}, \Delta z(\eta) \rangle - \beta \|P_{\text{keep}} \Delta z(\eta)\|. \tag{8}$$

The first term measures signed progress along the desired direction; the second measures the magnitude of changes in the subspace associated with already satisfied properties. A positive $\Delta_q(\eta)$ means that corrective progress outweighs weighted accompanying interference. This quantity is used for local task-representation analysis and does not directly imply a higher discrete checklist score; projected changes need not indicate damage to existing properties.

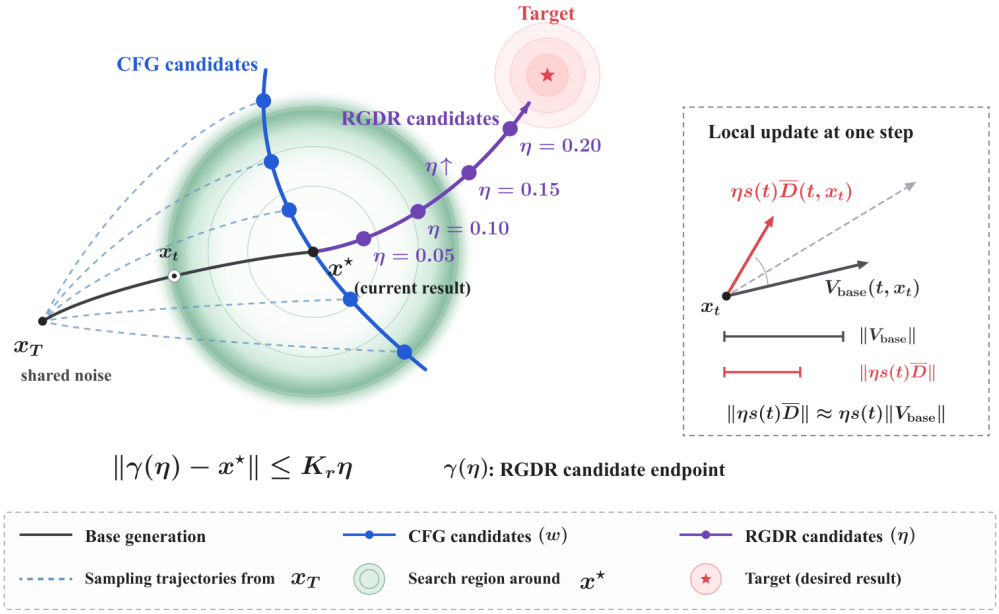

Figure 4: **Continuously controlled candidate space.** The blue and purple curves represent candidate endpoints parameterized by the CFG scale w and RGDR relative strength η.

Theorem 3.2 (Dominance of corrective progress induced by the feedback direction)**.** *Let $r > 0$. Suppose that z is twice continuously differentiable on $[0, r]$, with*

$$\sup_{\eta \in [0,r]} \|z''(\eta)\| \le M < \infty. \tag{9}$$

If the task response induced by D_F satisfies

$$\kappa_q(D_F) := \langle e_{\text{fix}}, z'(0^+) \rangle - \beta \|P_{\text{keep}} z'(0^+)\| > 0, \tag{10}$$

where $z'(0^+)$ is the right derivative at zero strength, then, for every $\eta \in [0, r]$,

$$\Delta_q(\eta) \ge \kappa_q(D_F)\eta - \frac{(1+\beta)M}{2}\eta^2. \tag{11}$$

Consequently, when $M > 0$, we have $\Delta_q(\eta) > 0$ for all strengths satisfying

$$0 < \eta < \min\left\{ r, \frac{2\kappa_q(D_F)}{(1+\beta)M} \right\}. \tag{12}$$

When $M = 0$, the same conclusion holds for all $0 < \eta \le r$.

The result permits accompanying interference, provided that first-order corrective progress at the reference exceeds its weighted contribution. The second-order response bound determines the strength interval over which this advantage holds. The proof and the dynamical expansion of the first-order response are provided in Appendices B.2 and B.3.

Target–Reference semantic construction. To emphasize the requested change and reduce unrelated intervention, we extract from feedback F two conditions describing the desired state, c_{target}, and the current erroneous state, $c_{\text{reference}}$. Using the frozen model's conditional velocity predictions V_θ, we construct

$$D_F(t, x) = V_\theta(t, x; c_{\text{target}}) - V_\theta(t, x; c_{\text{reference}}). \tag{13}$$

Both conditions focus on the attributes, objects, or relations to be corrected and retain the necessary context without restating unrelated requirements that are already satisfied. This construction aims to increase the first-order margin between corrective progress and interference in the first-order direction condition; it does not automatically satisfy that sufficient condition, and its effect is examined through generative responses and experiments.

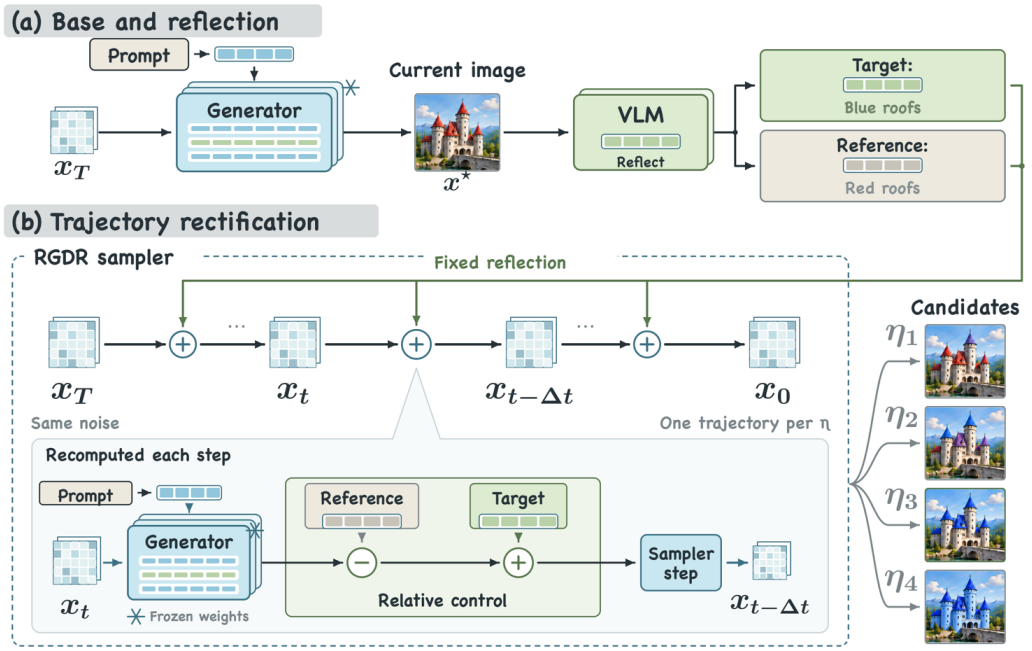

Figure 5: **RGDR pipeline overview.**

3.3 RGDR Algorithm

Combining the controlled superposition in Section 3.1 with the semantic construction in Section 3.2, RGDR derives Target and Reference conditions from reflections on a completed reference output. These describe the desired and observed states of the elements requiring correction, respectively, retaining relevant context without requesting changes to already-correct content. At each sampling step, RGDR recomputes the difference between the corresponding conditional velocity predictions, applies relative reparameterization, and adds the resulting correction to the base velocity field. With the generation configuration, feedback, and schedule fixed, candidates at different relative strengths are sampled separately from the same initial noise and evaluated alongside the reference for final selection. Model parameters remain fixed throughout. Algorithm 1 details the procedure, and Figure 5 illustrates the pipeline.

Algorithm 1 Reflection-Guided Dynamics Rectification (RGDR)

Input Task q; reference endpoint $x^\star$; fixed initial noise x_T;
base field V_{base}; frozen conditional field V_θ; feedback F; schedule s;
relative strength set $\mathcal{E}$; stabilizer $\varepsilon > 0$;
evaluator $\widehat{S}_{\text{sel}}(q, x)$ scoring the image corresponding to endpoint x

Output Selected endpoint $\widehat{x}$

Extract Target and Reference conditions from F
$D_F(t, x) \leftarrow V_\theta(t, x; c_{\text{target}}) - V_\theta(t, x; c_{\text{reference}})$
Evaluate both terms at the same (t, x)
$\overline{D}_F \leftarrow \|V_{\text{base}}\| D_F / (\|D_F\| + \varepsilon)$
$\mathcal{P} \leftarrow \{(0, x^\star)\}$
for all $\eta \in \mathcal{E}$ **do**
    Restart from the fixed initial noise x_T
    Integrate $V_{\text{base}} + \eta s(t) \overline{D}_F$ over the full interval from T to 0
Recompute the fields at each current state
    $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\eta, x_0)\}$
end for
$(\widehat{\eta}, \widehat{x}) \in \arg\max_{(\eta, x) \in \mathcal{P}} \widehat{S}_{\text{sel}}(q, x)$
return $\widehat{x}$

Table 1: **Comparison of follow-up rectification on PRISM-Bench and SciIR-Bench tasks.** Budget denotes the number of additional complete images per reference requiring correction. Bold indicates the best value in each column for a given model.

Method	Budget (images)	Final ↑ (0–100)	Gain ↑ (points)	Rel. gain ↑ (%)	Improved ↑ (%)	Repair ↑ (%)	Preserve ↑ (%)
Stable Diffusion 3.5 *Medium*							*Base* = 56.90
BoN	10	58.25	+1.35	2.38	12.0	4.11	96.14
TIR	10	58.14	+1.24	2.18	13.5	6.04	**98.39**
Dynamic CFG	10	59.05	+2.15	3.79	23.0	6.99	97.87
RGDR (Ours)	10	**60.64**	**+3.74**	**6.58**	**28.0**	**12.96**	96.44
SANA-1.5 1.6*B*							*Base* = 46.55
BoN	10	48.09	+1.54	3.31	13.0	3.40	**99.08**
TIR	10	49.96	+3.41	7.32	25.0	9.71	96.61
Dynamic CFG	10	48.97	+2.42	5.19	22.5	6.55	99.07
RGDR (Ours)	10	**50.92**	**+4.37**	**9.38**	**31.5**	**13.07**	94.46
FLUX.1-*dev*							*Base* = 70.30
BoN	10	71.57	+1.27	1.80	11.5	7.92	97.65
TIR	10	73.29	+2.99	4.25	15.5	10.23	**97.86**
Dynamic CFG	10	71.93	+1.63	2.32	16.5	8.78	97.65
RGDR (Ours)	10	**74.67**	**+4.36**	**6.21**	**26.5**	**19.15**	96.78
Qwen-Image-Edit-2511							*Base* = 72.77
BoN	5	74.84	+2.07	2.85	24.55	13.99	97.53
TIR	5	76.17	+3.41	4.68	30.91	20.32	97.25
Dynamic CFG	5	75.79	+3.03	4.17	29.09	16.27	**97.95**
RGDR (Ours)	5	**76.99**	**+4.23**	**5.81**	**33.64**	**23.58**	96.48

4 EXPERIMENTS

4.1 SETUP

To evaluate fine-grained rectification of verifiable residual errors, we build an evaluation set from PRISM-Bench and SciIR-Bench, covering compositional requirements and scientific image reasoning (Fang et al., 2025; Ma et al., 2026). References are generated by Stable Diffusion 3.5 Medium, SANA-1.5 1.6B, and FLUX.1-dev, or edited from SD3.5 Medium outputs by Qwen-Image-Edit-2511. All parameters remain frozen. Doubao Seed-2.1 Pro supplies feedback and scores outputs against fixed checklists on a 0–100 scale. We report Final, Gain, Rel. gain, and Improved for overall fulfillment and follow-up improvement, and Repair and Preserve for error correction and retention. Details are provided in Appendices C.1, C.2, and C.3.

4.2 MAIN EXPERIMENTS

Comparison of follow-up rectification methods. Following the evaluation protocol in Appendix C.3, we first perform BoN sampling under an initial image-generation budget for each task and model and select the highest-scoring candidate as the reference. Starting from these references, we compare continued BoN sampling (Ma et al., 2025), TIR prompt refinement (Khan et al., 2025), a guidance-scale baseline adapted from Dynamic CFG (Papalampidi et al., 2025), and RGDR across three text-to-image models and one image-editing model. All methods share the same reference and receive the matched additional-image budgets in Table 1. Reference construction and comparison protocols are detailed in Appendix C.4.

Table 1 shows that RGDR achieves the highest Final, Gain, Improved, and Repair across all four models. Averaged equally across models, its Gain exceeds that of BoN and TIR by 2.62 and 1.41 points, respectively. Under matched additional-image budgets, stepwise corrections along generation trajectories therefore realize more residual requirements than further noise or prompt adjustments. Figure 1 shows how targeted corrections can preserve the structure of existing scenes. In the bridal cycling example, RGDR adds the bicycle and riding action while retaining the side-view figure, wedding dress, and golden field, whereas TIR changes the subject's orientation and composition. In the text examples, modifications concentrate on the text region while retaining the

Table 2: **Correction direction ablation averaged across four models.** Gain is measured in score points and Repair in percent. Bold indicates the best value in each column.

	Category-wise Gain ↑			Overall	
Direction	Human	Open	Scientific & Text	Gain ↑	Repair (%) ↑
TR Correct	**7.56**	**5.71**	**3.18**	**4.34**	**15.28**
Operation	2.09	2.75	2.65	2.60	7.95
TR Wrong	3.87	1.56	0.45	1.15	4.36
Gaussian	0.37	0.43	1.82	1.31	4.16

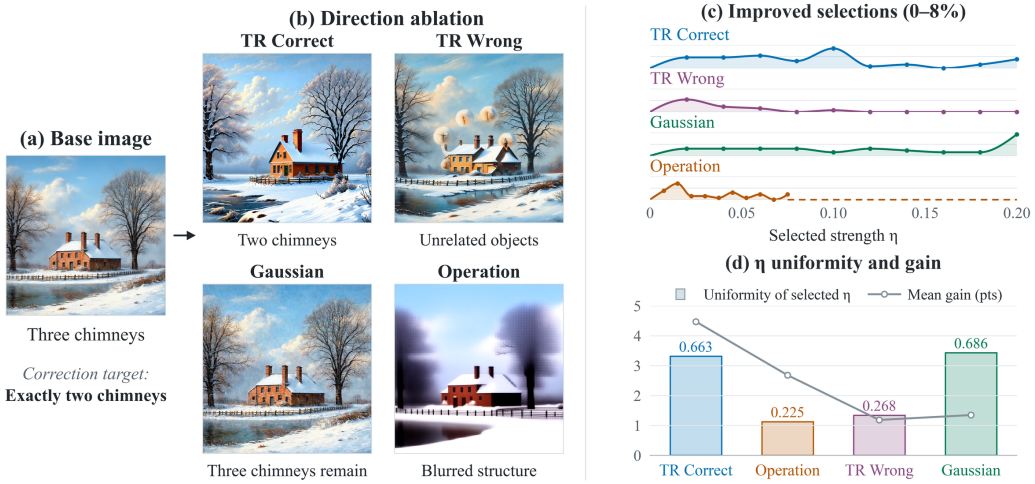

Figure 6: **Qualitative comparison of correction directions and strength analysis on SD3.5 Medium.** Panels (a,b) show a qualitative example under different correction directions. Panels (c,d) report the selected-strength distributions of improved outputs, selected-strength uniformity, and mean gain on a separate paired visualization set.

surrounding border or sign setting. These comparisons address the difficulty of realizing requested changes while controlling visual variation illustrated in Figure 2. RGDR also achieves 1.81 times the mean Gain of the adapted Dynamic CFG baseline, supporting the error-specific additional direction illustrated in Figure 4. Reflection-guided trajectory correction thus provides greater rectification gains than adjusting native guidance strength alone. The same advantage on previously edited outputs shows that RGDR can serve as an additional rectification mechanism within existing generation and editing workflows.

4.3 Ablation Studies

Correction direction construction. Why use Target–Reference to construct the correction direction D_F in RGDR? Motivated by the analysis of corrective progress and accompanying interference in Theorem 3.2, we compare TR Correct, Operation, TR Wrong, and Gaussian on the rectification subsets of all four models. All configurations share references, relative reparameterization, and the evaluation protocol. The constructions of these four methods are detailed in Appendix C.5.

TR Correct achieves the highest Gain in all three task categories in Table 2, with an overall Gain of 4.34 points and Repair of 15.28%. The SD3.5 Medium chimney example in Figure 6 illustrates the semantic effects of different directions. TR Correct changes three chimneys to two; Gaussian mainly alters minor details while leaving the incorrect count unchanged; and TR Wrong introduces unrelated objects. On the separate paired visualization set for SD3.5 Medium in Figure 6(c,d), improved selections for TR Correct span multiple strengths from weak interventions up to $\eta = 0.20$, while achieving the highest mean Gain. Gaussian also has broadly distributed selected strengths, but its mean Gain is only about one-third of that of TR Correct. Improved selections for Operation concentrate at small strengths, whereas TR Wrong yields none above $\eta = 0.10$. These results show that the value of a direction lies not in larger perturbations alone, but in producing useful,

target-specific corrections at different intervention strengths. TR Correct provides improved candidates across a broad range of strengths while achieving higher rectification gains, empirically supporting the Target–Reference design motivated by Theorem 3.2.

5 Conclusion

We introduce Reflection-Guided Dynamics Rectification (RGDR), a training-free method that translates reflections on existing images into corrections along sampling trajectories to address residual errors in generated outputs. Under suitable conditions, we prove that candidate outputs vary continuously with correction strength and that their deviation from the original output is bounded. We also establish a sufficient condition under which local corrective progress outweighs weighted interference. Experiments show that reflection-guided interventions along sampling trajectories reduce residual errors in generated images. In our ablations, the Target–Reference construction is more effective when modification requests are limited to erroneous content, consistent with the sufficient condition established above. This framework could be extended to a broader range of models and generation tasks.

AI Use Statement

We used generative AI tools to assist with manuscript polishing, the writing of mathematical proofs, the refinement of existing hypotheses, code development, and figure generation. We did not use these tools to originate the core research ideas or claims. The use of generative Al tools to generate synthetic datasets is not applicable to this work, In particular, AI-assisted proofs and hypothesis refinements were checked for logical consistency, code was reviewed and verified for correctness, and figures were inspected for accuracy and scientific plausibility. We take responsibility for the final content of this work, including all text, claims, code, and figures produced with the assistance of generative AI tools.

REPRODUCIBILITY STATEMENT

To make our results reproducible, we provide relevant details in the appendix, including theoretical proofs, task and checklist construction, model and sampling configurations, evaluation protocols, and implementation details of the proposed method.

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A Extended Related Work

Noise optimization and multi-round reflection. Beyond the noise sampling and prompt refinement methods discussed in the main text, ReNO (Eyring et al., 2024) optimizes initial noise using reward gradients. Idea2Img (Yang et al., 2023) iteratively updates prompts based on diagnoses of generated images, while PRIS (Kim et al., 2025) aggregates recurring errors across candidates to guide prompt redesign. Other work enhances multi-round reflection through training. SRRL (Pan et al., 2025) and RL-RIG (Wang et al., 2026b) use reinforcement learning to improve multi-round image generation and reflection-based editing, respectively.

Particle and tree search along generation trajectories. Feynman–Kac Steering (Singhal et al., 2025) explores generation trajectories through particle resampling, whereas Classical Search (Zhang et al., 2025) combines local and tree search to expand sampling paths. These methods complement the trajectory selection, guidance-scale adjustment, and semantic diagnostic interventions discussed in the main text.

B Mathematical Proofs

We retain the definitions and assumptions of the main text. Vector norms are Euclidean, and matrix norms are the corresponding induced operator norms.

B.1 Proof of Theorem 3.1

Let L_V, L_D be the uniform state-Lipschitz constants of the base and direction fields, respectively, and choose a finite $C_D \geq 0$ such that $\|D(t,x)\| \leq C_D$. For $\alpha, \alpha' \in [0, r]$, let $x_t^{\alpha}, x_t^{\alpha'}$ be the solution trajectories starting from the same initial noise x_T, and write $e(t) = \|x_t^{\alpha} - x_t^{\alpha'}\|$. Their integral equations and the Lipschitz assumptions give

$$e(t) \leq \int_t^T \left[(L_V + rL_D)e(u) + C_D|\alpha - \alpha'|\right] \mathrm{d}u. \tag{B.1}$$

Set $L_r = L_V + rL_D$. Applying Grönwall's inequality (Teschl, 2012, Lemma 2.7) after the change of time variable $\tau = T - t$ yields

$$e(t) \leq C_D|\alpha - \alpha'| \int_t^T e^{L_r(u-t)} \mathrm{d}u. \tag{B.2}$$

Taking $t = 0$ and defining $K_r = C_D \int_0^T e^{L_r u} \mathrm{d}u < \infty$, we obtain

$$\|\gamma_D(\alpha) - \gamma_D(\alpha')\| \leq K_r|\alpha - \alpha'|. \tag{B.3}$$

Setting $\alpha' = 0$ and using $x^\star = \gamma_D(0)$ then gives $\|\gamma_D(\alpha) - x^\star\| \leq K_r\alpha$. This completes the proof. The integral expression for K_r also covers $L_r = 0$ and $C_D = 0$.

Application after relative reparameterization. Suppose additionally that $\|V_{\text{base}}(t,x)\| \leq C_V < \infty$ on the common trajectory region. The reverse triangle inequality and the $2/\varepsilon$-Lipschitz bound for $u \mapsto u/(\|u\| + \varepsilon)$ imply that the reparameterized direction in the main text satisfies

$$\begin{aligned} \|\overline{D}(t,x)\| &\leq C_V, \\ \|\overline{D}(t,x) - \overline{D}(t,y)\| &\leq \left(L_V + \frac{2C_V L_D}{\varepsilon}\right) \|x - y\|. \end{aligned} \tag{B.4}$$

Thus, for a fixed continuous bounded schedule s, the field $s\overline{D}$ satisfies the same types of boundedness and state-Lipschitz conditions, and the preceding comparison applies to $\gamma(\eta)$. The same integral estimate also holds for bounded piecewise-continuous schedules, provided the corresponding solutions exist.

B.2 Proof of Theorem 3.2

Let $w = z'(0^+)$. Applying Taylor's formula with an integral remainder componentwise to z (Knapp, 2016, Theorem 1.36) gives

$$\Delta z(\eta) = \eta w + R(\eta), \qquad R(\eta) = \int_0^\eta (\eta - u) z''(u)\,\mathrm{d}u, \qquad \|R(\eta)\| \le \frac{M}{2}\eta^2. \tag{B.5}$$

Using $\|e_{\mathrm{fix}}\| = 1$, $\|P_{\mathrm{keep}}\| \le 1$, and the triangle inequality,

$$\begin{aligned} \Delta_q(\eta) &= \langle e_{\mathrm{fix}}, \eta w + R(\eta)\rangle - \beta \|P_{\mathrm{keep}}(\eta w + R(\eta))\| \\ &\ge \eta\big(\langle e_{\mathrm{fix}}, w\rangle - \beta\|P_{\mathrm{keep}} w\|\big) - (1+\beta)\|R(\eta)\| \\ &\ge \kappa_q(D_F)\eta - \frac{(1+\beta)M}{2}\eta^2. \end{aligned} \tag{B.6}$$

When $\kappa_q(D_F) > 0$ and $M > 0$, the right-hand side is strictly positive for

$$0 < \eta < \min\left\{ r, \frac{2\kappa_q(D_F)}{(1+\beta)M} \right\} \tag{B.7}$$

When $M = 0$, we have $R(\eta) = 0$, so $\Delta_q(\eta) = \kappa_q(D_F)\eta > 0$ for every $0 < \eta \le r$. This completes the proof.

B.3 Dynamical Expression of the First-Order Response

To expand $z'(0^+)$ in the main text, additionally assume that V_{base} and $s\overline{D}_F$ are continuously differentiable in the state, with state derivatives bounded in a neighborhood of the relevant trajectories, and that Φ_q is differentiable at $x^\star$. The fields and their state derivatives are continuous in time. Under these conditions, differentiable dependence on parameters (Teschl, 2012, Theorem 9.16) justifies differentiating the integral equation with respect to η. Let x_t^0 denote the reference trajectory and define

$$p_t = \partial_\eta x_t^\eta|_{\eta=0^+}, \qquad A(t) = \mathrm{D}_x V_{\mathrm{base}}(t, x_t^0). \tag{B.8}$$

Differentiating the sampling dynamics with respect to η and evaluating at $\eta = 0$ yields

$$\frac{\mathrm{d}p_t}{\mathrm{d}t} = A(t)p_t + s(t)\overline{D}_F(t, x_t^0), \qquad p_T = 0. \tag{B.9}$$

The condition $p_T = 0$ follows from using the same initial noise at every strength. Let the state-transition matrix $\Psi(t,u)$ satisfy $\partial_t \Psi(t,u) = A(t)\Psi(t,u)$ and $\Psi(u,u) = I_d$. The variation-of-constants formula (Teschl, 2012, Theorem 3.12), followed by the chain rule, yields

$$\begin{aligned} p_0 &= \int_T^0 \Psi(0,t)s(t)\overline{D}_F(t, x_t^0)\,\mathrm{d}t, \\ z'(0^+) &= \mathrm{D}\Phi_q(x^\star)\int_T^0 \Psi(0,t)s(t)\overline{D}_F(t, x_t^0)\,\mathrm{d}t. \end{aligned} \tag{B.10}$$

This expression propagates each step's correction through the subsequent dynamics to the generated endpoint and then maps it into the task representation. The integration direction agrees with the $T \to 0$ convention in the main text. The additional smoothness assumptions in this subsection are used for this dynamical expansion, not for the preceding proof stated directly in terms of z.

C Experimental Details

C.1 Evaluation Set and Task Library

We organize a task library from PRISM-Bench and SciIR-Bench. To describe its content coverage, we organize tasks into three groups.

1. **Human.** This group covers human-centered images and requirements concerning pose, hand gestures, and interactions.
2. **Open.** This group covers general scenes and requirements concerning objects, attributes, spatial relations, and environmental conditions.
3. **Scientific & Text.** This group combines scientific diagrams with text-containing images, including scientific structures and relations, diagram labels, and specified rendered text.

The task library contains 1,500 entries, comprising 700 from PRISM-Bench and 800 from SciIR-Bench. Following the three groups defined above, it includes 176 Human tasks, 384 Open tasks, and 940 Scientific & Text tasks. Each entry includes the original prompt, source information, and its associated checklist. For PRISM-Bench, we construct and review task-specific checklists from the original prompts to assess the requested visual content. For SciIR-Bench, we retain the officially provided checklist questions and categories.

From this library, we curate a 300-task evaluation set containing 100 Human tasks, 100 Open tasks, and 100 Scientific & Text tasks. The main experiments, direction ablations, and temporal-scheduling experiments use their respective task subsets from the library, whose sizes need not match this evaluation set. Within each comparison, methods share the same tasks, reference outputs, and checklists. The reported results use the task subsets and checklist versions of the original evaluation runs.

C.2 Model Configuration

Table S1 lists the sampling configurations. Inference uses BF16 precision and a resolution of 1024×1024. The three text-to-image models use their native guidance scales, whereas Qwen-Image-Edit-2511 uses true CFG scale. RGDR uses $s(t_j) = j/(N-1)$ for SD3.5 Medium and $s(t_j) = 1$ for the other three models.

Table S1: Sampling and guidance configurations. $j = 0, \ldots, N-1$ indexes the executed sampling steps.

Model	Steps N	Guidance scale	RGDR schedule $s(t_j)$
SD3.5 Medium	40	4.5	$j/39$
SANA-1.5 1.6B	20	4.5	1
FLUX.1-dev	50	3.5	1
Qwen-Image-Edit-2511	40	4.0	1

C.3 Evaluation, Selection, and Aggregation

We evaluate images against fixed task-specific checklists and select final outputs under the same evaluation protocol. Let task i have K_i checklist items, and let $a_{ik}(I) \in [0,1]$ be the numerical score of image I on item k. The task score is

$$S_i(I) = \frac{100}{K_i} \sum_{k=1}^{K_i} a_{ik}(I). \tag{C.1}$$

Let b_i, f_i denote the scores of the shared reference and selected output, respectively, and let n be the number of tasks in the aggregate. We report

$$\begin{aligned} \text{Final} &= \frac{1}{n}\sum_i f_i, & \text{Gain} &= \frac{1}{n}\sum_i (f_i - b_i), \\ \text{Rel. gain} &= 100\frac{\text{Gain}}{\frac{1}{n}\sum_i b_i}, & \text{Improved} &= \frac{100}{n}\sum_i \mathbf{1}\{f_i > b_i + \delta\}. \end{aligned} \tag{C.2}$$

The recorded protocols for the three generative models use an improvement tolerance of 10^{-10} on the $[0, 1]$ scale, equivalent to $\delta = 10^{-8}$ on the 0–100 scale. Rel. gain is the ratio of mean gain to mean reference score, not the mean of per-task relative percentages.

Repair and Preserve use paired pass/fail judgments. For task i, let $N_{ab}^{(i)}$ count items with reference judgment a and output judgment b, where 0 denotes fail and 1 denotes pass. Items marked uncertain at either endpoint are excluded. Define $\mathcal{I}_R = \{i : N_{00}^{(i)} + N_{01}^{(i)} > 0\}$ and $\mathcal{I}_P = \{i : N_{10}^{(i)} + N_{11}^{(i)} > 0\}$. Then

$$\begin{aligned} \text{Repair} &= \frac{100}{|\mathcal{I}_R|} \sum_{i \in \mathcal{I}_R} \frac{N_{01}^{(i)}}{N_{00}^{(i)} + N_{01}^{(i)}}, \\ \text{Preserve} &= \frac{100}{|\mathcal{I}_P|} \sum_{i \in \mathcal{I}_P} \frac{N_{11}^{(i)}}{N_{10}^{(i)} + N_{11}^{(i)}}. \end{aligned} \tag{C.3}$$

Each metric is macro-averaged over tasks with a nonzero denominator. Numerical scores and binary judgments are recorded separately. An uncertain item can contribute its numerical score to S_i, but not to Repair or Preserve. Results are pooled across sources using the relevant valid-task sets; cross-model comparisons assign equal weight to each model.

C.4 Main-Experiment Comparison Protocol

For Stable Diffusion 3.5 Medium, SANA-1.5 1.6B, and FLUX.1-dev, we generate 20 candidate images under the original prompt using different noise initializations. We score these candidates according to Appendix C.3 and select the highest-scoring image as the Best@20 reference. For Qwen-Image-Edit-2511, we use an SD3.5 Medium output as the fixed source image, generate five editing candidates, and select the highest-scoring candidate as the Best@5 reference using the same scoring protocol.

Starting from these references, we compare continued BoN sampling, TIR, the guidance-scale baseline adapted from Dynamic CFG, and RGDR. Each method receives an additional budget of ten complete images for each text-to-image model and five for the image-editing model. Tasks and checklists remain fixed across methods. For each method, we score the additional candidates using the same protocol and select the highest-scoring image from the reference and all additional candidates as the final output.

BoN. Following BoN candidate generation and selection(Ma et al., 2025), we retain the original prompt or editing instruction, generate candidates from different initial noise, and select by checklist score.

TIR. The prompt or editing instruction is revised iteratively using the current image and its evaluation feedback(Khan et al., 2025). The reference seed remains fixed. Each round produces a complete candidate under the revised condition, which informs the next reflection.

Dynamic CFG. We use a constant-schedule adaptation within the guidance-schedule space of Dynamic CFG(Papalampidi et al., 2025), imposing the same guidance scale at every step of a trajectory, $w_j \equiv w$. With the prompt and reference noise fixed, complete candidates are generated for different values of w and then evaluated for selection. SD3.5 Medium and SANA use ten uniformly spaced guidance values in $[2.5, 6.5]$; FLUX uses ten uniformly spaced native guidance-parameter values in $[2.1, 4.9]$. For Qwen-Image-Edit-2511, we use $w \in \{1.5, 2.5, 3.5, 5.0, 6.0\}$, generating one complete candidate per value from the same fixed original input image.

RGDR. Target–Reference conditions and reference noise remain fixed, and complete candidates are generated independently at different relative strengths. The text-to-image models use $\mathcal{E} = \{0.02, 0.04, \ldots, 0.20\}$ and the schedules in Table S1. The additional correction field is recomputed at each sampling step. For Qwen-Image-Edit-2511, we use $\eta \in \{0.01, 0.025, 0.05, 0.075, 0.10\}$ with constant scheduling, generating one complete candidate per value from the same fixed original input image.

C.5 Ablation Directions and Conditioning Texts

We construct conditioning texts and correction directions based on the task requirements and residual errors in the reference image. Target and Reference describe contrasting states associated with the same correction, whereas Operation expresses the correction as a direct instruction.

Target. The Target describes the desired state after the specified error is corrected. It identifies the required attributes, objects, or relations, retaining the context needed to locate the change without restating unrelated requirements already satisfied.

Reference. The Reference describes the current erroneous state associated with the same correction target. It contrasts with the Target around the same objects, attributes, or relations.

Operation. The same correction request is expressed as a direct operation-style instruction, including the content to preserve. This ablation directly uses the model prediction conditioned on that instruction as the additional direction, without a Target–Reference prediction difference.

Gaussian. This ablation uses a random direction without task semantics.

TR Correct is constructed by subtracting the conditional velocity prediction under Reference from that under Target, with both descriptions conditioning the frozen model. Within each rectification attempt, the two texts remain fixed, while their conditional velocity predictions are recomputed at the current state and time. TR Wrong retains the Target–Reference difference formulation but uses mismatched feedback to construct the correction direction. All directions use relative reparameterization.

D Temporal Scheduling Experiments

D.1 Schedule Functions and Comparison Setting

We compare constant, increasing, and decreasing correction-strength schedules on three text-to-image models, Stable Diffusion 3.5 Medium, SANA-1.5 1.6B, and FLUX.1-dev. Let N denote the number of sampling steps and $j = 0, \ldots, N-1$ the execution-step index. For relative strength η, the coefficient applied at step j is $\eta_j = \eta s(t_j)$, where

$$s_{\text{const}}(t_j) = 1, \qquad s_{\text{inc}}(t_j) = \frac{j}{N-1}, \qquad s_{\text{dec}}(t_j) = 1 - \frac{j}{N-1}. \tag{D.1}$$

For each model, the three schedules are compared on the same task subset using shared Best@20 references, initial noise, and Target–Reference conditions. We generate candidates using the same peak-strength grid $\eta \in \{0.02, 0.04, \ldots, 0.20\}$ and evaluate and select outputs according to Appendix C.3. Model configurations follow Appendix C.2.

D.2 Results and Analysis

Table S2 shows that all three schedules yield positive rectification gains, but their ranking differs across models. For SD3.5 Medium, the increasing schedule achieves a Final score of 62.65 and a Gain of 3.69, compared with gains of 2.81 for decreasing and 2.32 for constant scheduling. Increasing the correction strength throughout sampling therefore improves Gain over constant scheduling by 1.37 points on this task subset.

SANA and FLUX show a different ordering, with constant scheduling performing best, followed by decreasing and then increasing scheduling. For SANA, constant scheduling achieves a Gain of 5.33, exceeding increasing scheduling by 1.93 points. For FLUX, the corresponding gains are 4.68 and 2.99, a difference of 1.69 points. Decreasing scheduling remains effective on both models, yielding gains of 4.81 and 3.79, respectively. Thus, no single schedule is best across all three models. These comparisons show that the temporal distribution of correction strength affects rectification performance, supporting model-specific scheduling choices.

Table S2: Temporal-scheduling results. Reference is the mean reference score. Final and Gain are reported in score points. Const., Inc., and Dec. denote constant, increasing, and decreasing schedules. Each row compares schedules on the same tasks. Bold indicates the best value within each row and metric.

		Final			Gain		
Model	Reference	Const.	Inc.	Dec.	Const.	Inc.	Dec.
SD3.5 Medium	58.96	61.28	**62.65**	61.77	2.32	**3.69**	2.81
SANA-1.5 1.6B	50.39	**55.72**	53.79	55.20	**5.33**	3.40	4.81
FLUX.1-dev	68.07	**72.75**	71.06	71.86	**4.68**	2.99	3.79

E SUPPLEMENTARY QUALITATIVE RESULTS

This appendix presents 24 additional qualitative examples in four categories: Human, Open, Text, and Scientific. Each pair compares the reference output (Base, left) with a selected RGDR result (right); the description below identifies the correction target for the corresponding pair.

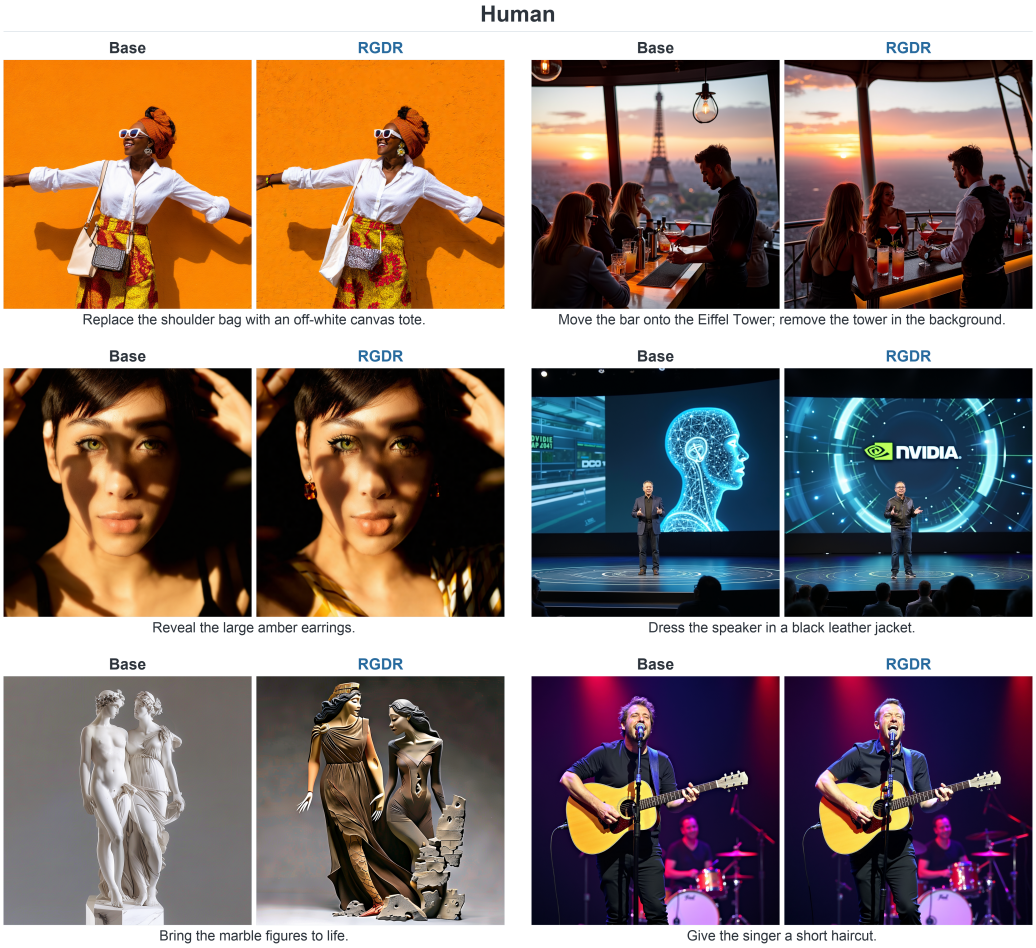

Figure 7: **Human.** Base (left) and RGDR (right).

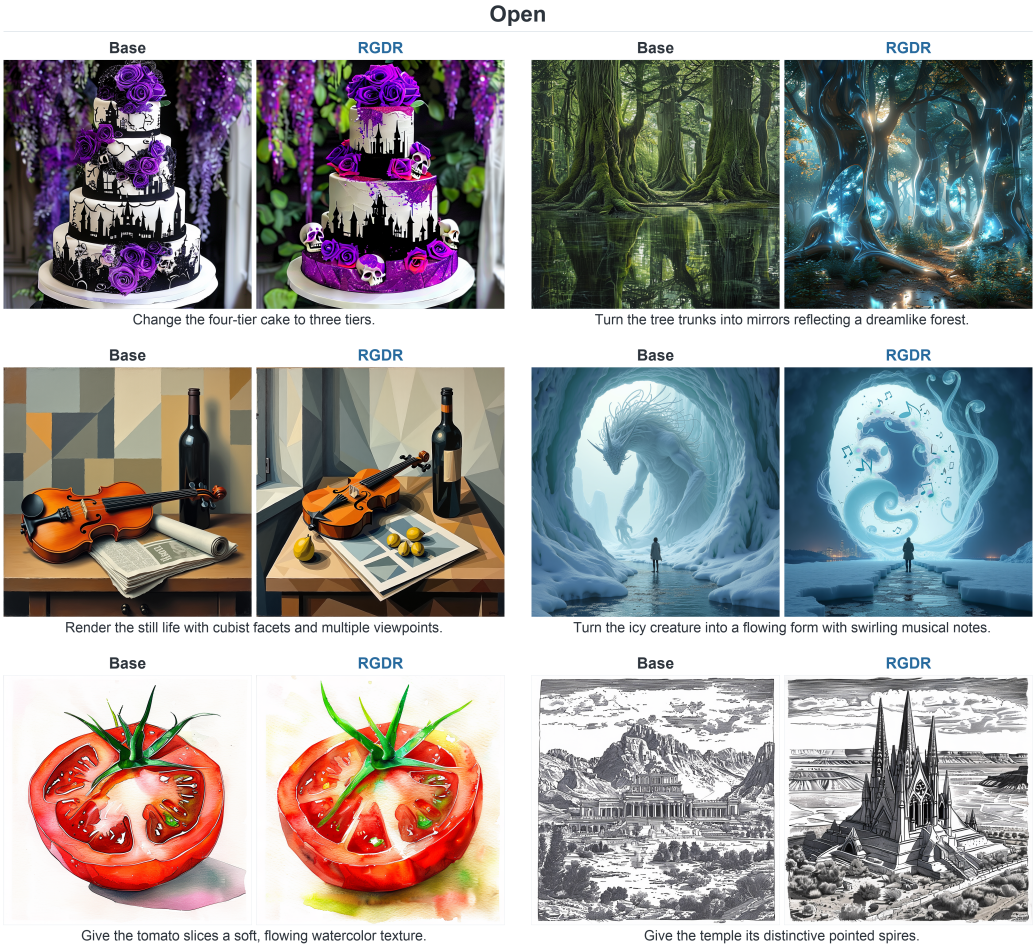

Figure 8: **Open.** Base (left) and RGDR (right).

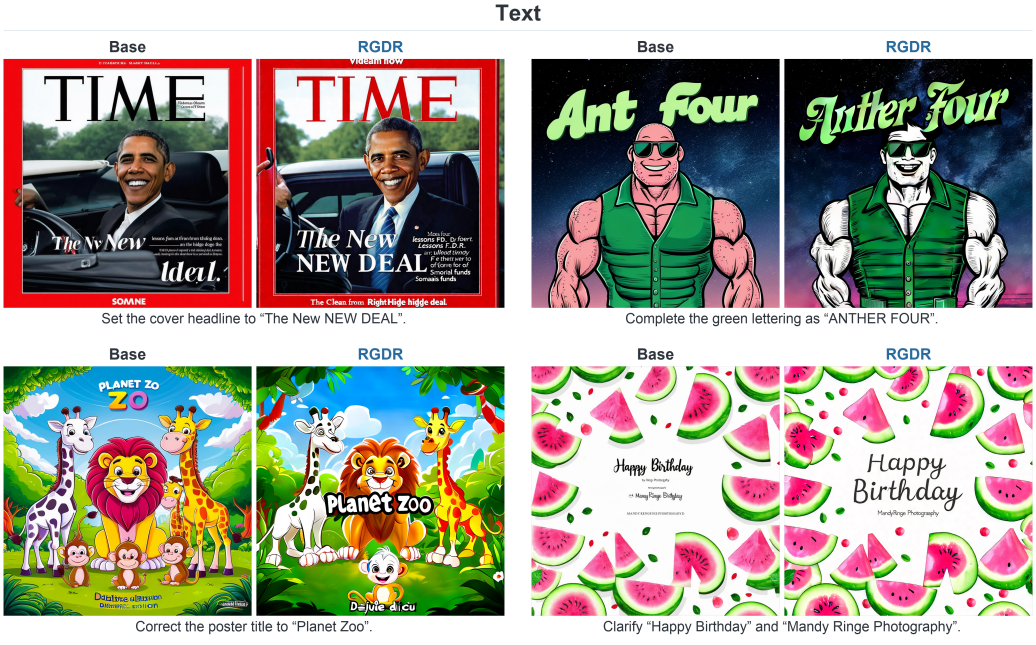

Figure 9: **Text.** Base (left) and RGDR (right).

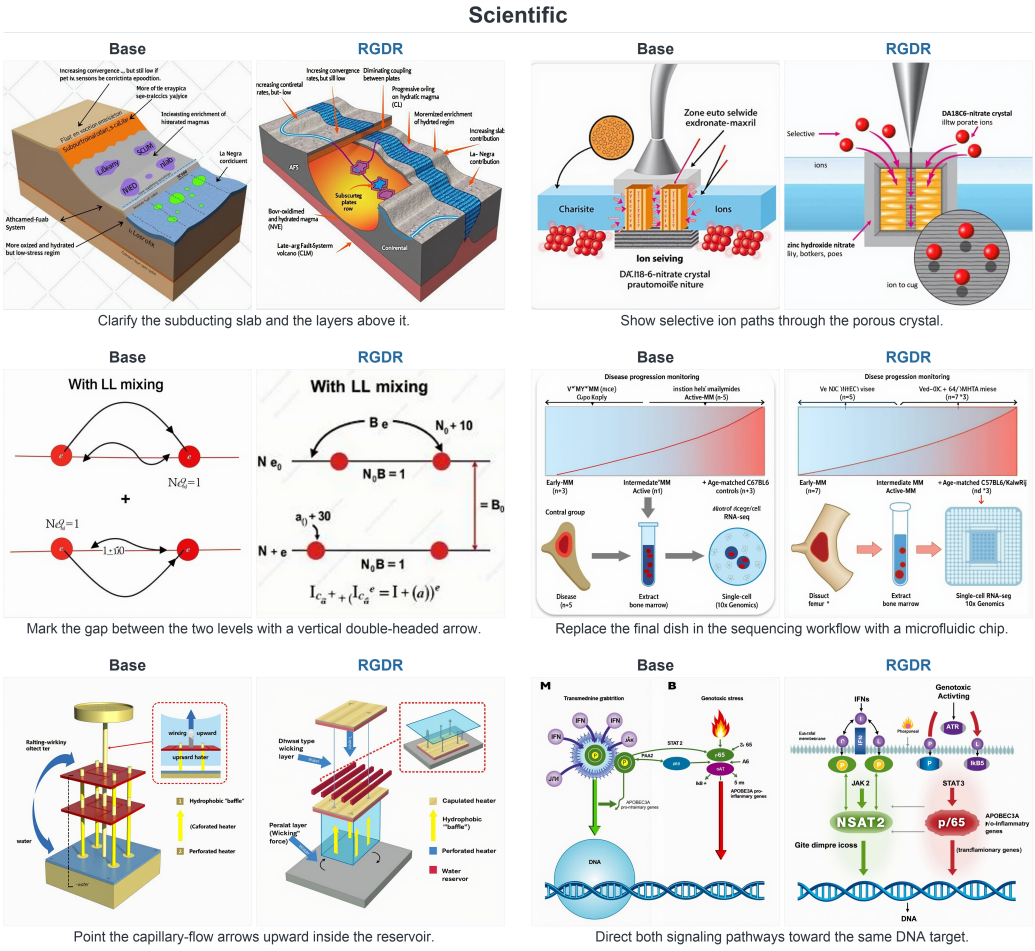

Figure 10: **Scientific.** Base (left) and RGDR (right).